

第六次组会记录

Part I. Review

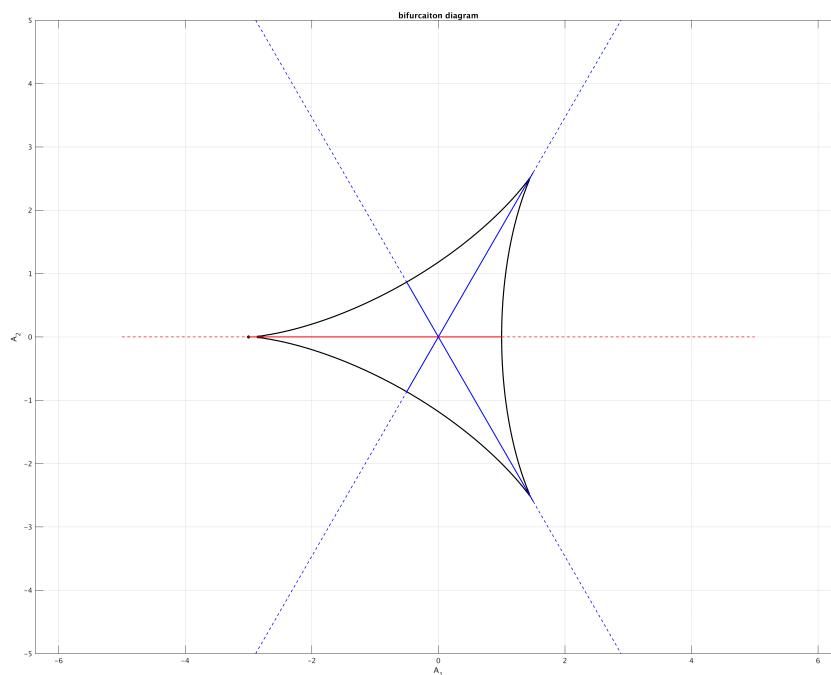
bifurcation diagram in the case : $B_1 = 1, B_2 = 0$

Last time we considered $B_1 = 1, B_2 = 0$, this gives us :

$$H(x, y) = -\frac{1}{2}(x^2 + y^2) - A_2x^3 - A_2xy^2 + (1 - A_1)x^2y - \frac{3A_1 + 1}{3}y^3$$

The bifurcation diagram is :

$$\begin{cases} \Sigma_1 : & -27 + 18A_1^2 + 8A_1^3 + A_1^4 + 18A_2^2 - 24A_1A_2^2 + 2A_1^2A_2^2 + A_2^4 = 0 \\ \Sigma_2 : & \{A_2 = 0 \text{ with } -3 < A_1 < 1\} \cup \{3A_1^2 - A_2^2 = 0 \text{ with } -\frac{1}{2} < A_1 < \frac{3}{2}\} \end{cases} \quad (*)$$



cubic form , discriminant and multiple root

Definition : (*cubic form*)

A real cubic form is like :

$$Q(x, y) = ax^3 + bx^2y + cxy^2 + dy^3$$

Where $a, b, c, d \in \mathbb{R}$

Property of cubic form :

The following is equivalent when $y \neq 0$:

$$Q(x, y) = 0 \iff at^3 + bt^2 + ct + d = 0$$

where $t := \frac{x}{y}$

Definition : (Σ_3)

Given a cubic form with parameter $\lambda := (\alpha_1, \alpha_2, \dots)$:

$$Q(x, y) = a(\lambda)x^3 + b(\lambda)x^2y + c(\lambda)xy^2 + d(\lambda)y^3$$

$\Sigma_3 := \{\lambda : \text{two lines in } Q(x, y) = 0 \text{ coincide}\}$

Remark :

you can look it as : $at^3 + bt^2 + ct + d$ has multiple root with $t = \frac{x}{y}$

Discriminant in cubic :

In quadratic case we know that the discriminant is $\Delta = b^2 - 4ac$.

Actually in cubic case, we also have discriminant as following :

Given $q(t) = at^3 + bt^2 + ct + d = 0$

The discriminant of q is :

$$\Delta = 18abcd - 4b^3d + b^2c^2 - 4ac^3 - 27a^2d^2$$

We have the following table about roots :

Discriminant	How many roots in $q(t)$	How many real lines in $Q(x, y)$
$\Delta > 0$	3 different real roots	3 different real lines
$\Delta = 0$	has multiple roots	2 or 1 different real lines
$\Delta < 0$	1 real root, a pair of conjugate complex roots	1 real line

Part II. Problem

Determine how the *phase portrait* looks like in each component of bifurcation diagram (*).